INDUSTRIAL ECONOMICS and MANAGEMENT (ME 491)

SCOPE

- ✓Management, Managerial Planning and Decision Making
- ✓Organising, Leading and Controlling
- ✓Fundamentals of Accounting and Costing
- ✓Engineering Economy
- ✓Project Management

COURSE OBJECTIVES

- Introduce students to the basics of management, managerial planning and the decision making process.
- To equip students with the tools and techniques for planning and decision making.
- To equip students with the knowledge and techniques:
  - ✓ To assess the economic viability of alternative solutions to technical problems and propose solutions.
  - ✓To justify and defend the solutions proposed by them.
- Understand the economic consequences of decision making regarding savings, investment, etc and the role that financial planning and economic optimization play in them.
- Understand effective project management

COURSE EXECUTION

Lecturers: Dr L. D. Mensah (Mechanical Engineering Dept.)

Mrs A. K. Intsiful (Mechanical Engineering Dept.)

Mr. S. H. M. Aikins (Agricultural Engineering Dept.)

Lecture Periods: Tuesdays: 8am – 11 am (TE IV & Biomedical Eng IV)

Tuesday: 12noon – 3 pm (EE IV & Comp.Eng. IV)

Wednesdays: 1 – 3 pm (Mech. IV, Agric. IV & Mat. Eng. IV)

Consultation times: Mondays & Wednesdays – 8.30am – 10 am.

Thursdays – 10.30 am – 1pm

Or

By Appointment (for both lecturers)

Recommended Books For Further Reading

1. Fundamentals of Management by Donnelly, Gibson and Ivancevich **or any book on Management.**
2. 2. Engineering Economy by L. T. Blank and A.J. Tarquin, McGraw-Hill Press, 4th edition, ISBN 0-07-063110-7 **or any book on Engineering Economy**
3. Accounting, Principles and Application or **any basic Accounting book**.
4. Principles of Operations Management by Render and Heizer **or any book on P O M. or Operations Management**

COURSE BREAKDOWN - 1

- **GENERAL PRINCIPLES OF MANAGEMENT** **4 wks**

Introduction to management Aug. 28th & 29
Planning and decision making, Forecasting Aug. 28 & 29.
Controlling Sept 4th & 5th
Organizing and staffing Sept. 11 &12
Leading Sept. 18 & 19

- **FINANCIAL MANAGEMENT** **2 wks.**

Accounting, Financial Reports and business ratios Sept. 25th & 26th
Cost Accounting Oct. 2nd & 3rd

- **ENGINEERING ECONOMY** **3 wks**

Basic concepts and graphics, factors and their use, Oct. 9th & 10th
nominal and effective interest rates, rate of return Oct. 16th & 17th
computations, benefit cost ratio evaluation, Oct. 23rd & 24th
break even analysis and payback period,
determining MARR etc

COURSE BREAKDOWN - 2

- **PROJECT MANAGEMENT** **2 - 3 wks**

Characteristics of a Project, Data collection and	Oct.	30th & 31st
Analysis, Project Scheduling- Network based	Nov.	6th & 7th
Methods, Implementation and Control	Nov.	13th & 14th

RESPONSIBILITIES OF STUDENTS

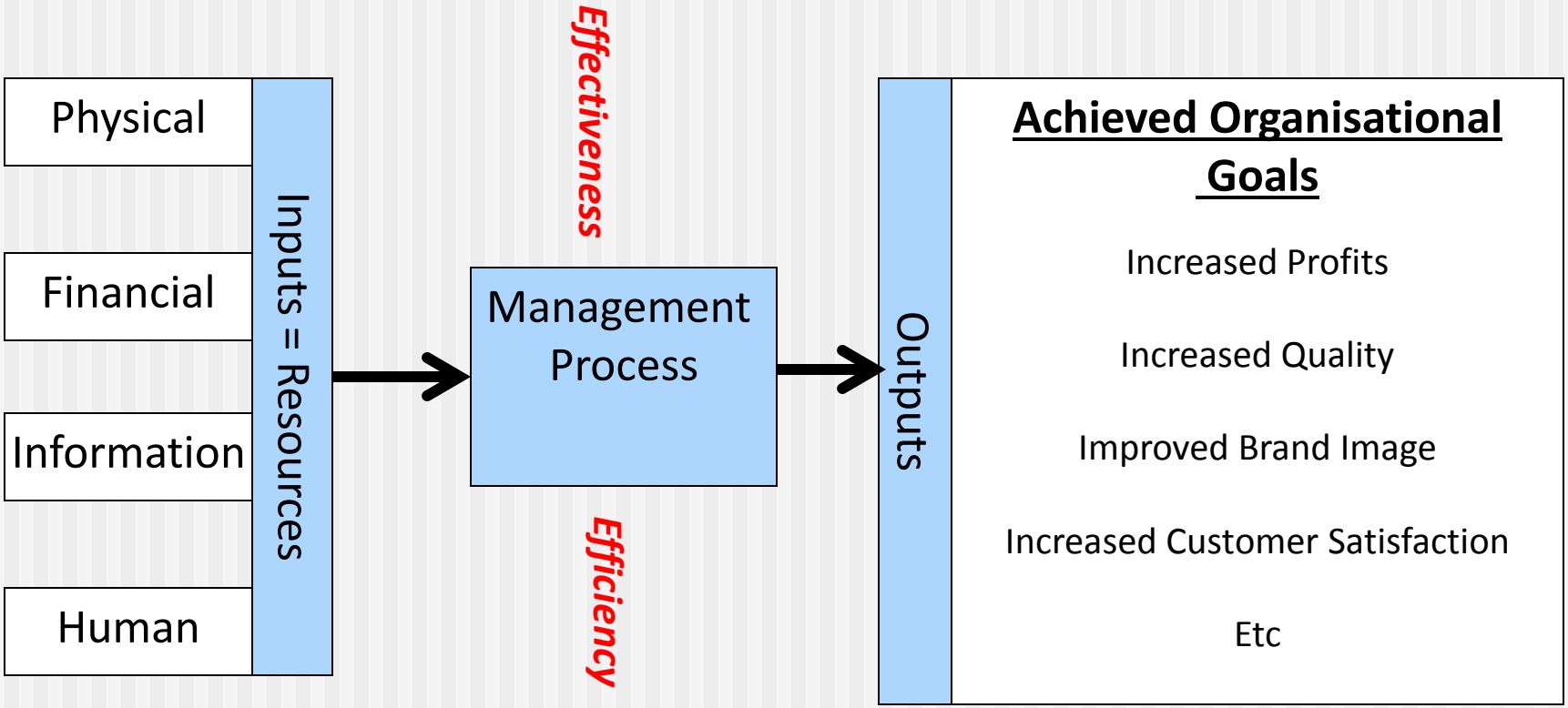
All students have to acquire all materials given in class and will be held accountable for all work that is assigned.

- **Continuous Assessment**: Mid semester exam - 30%
- **Final Exams**: 70%

PRINCIPLES OF MANAGEMENT (1)

- **Management** is the process of *coordinating* and *utilizing* human and nonhuman *resources* in pursuit of organizational profit, service, and survival *goals*.
- Management is a form of work which involves *coordinating* an organisation's *resources* to achieve a *goal*.
- Management can be defined as a set of activities directed at the *efficient* and *effective* utilization of *resources* in the pursuit of one or more *goals*.

PRINCIPLES OF MANAGEMENT (2)



Effectiveness: to achieve organisational goals
Efficiency: To achieve organisational goals with minimum wastage of resources

MANAGEMENT PROCESS



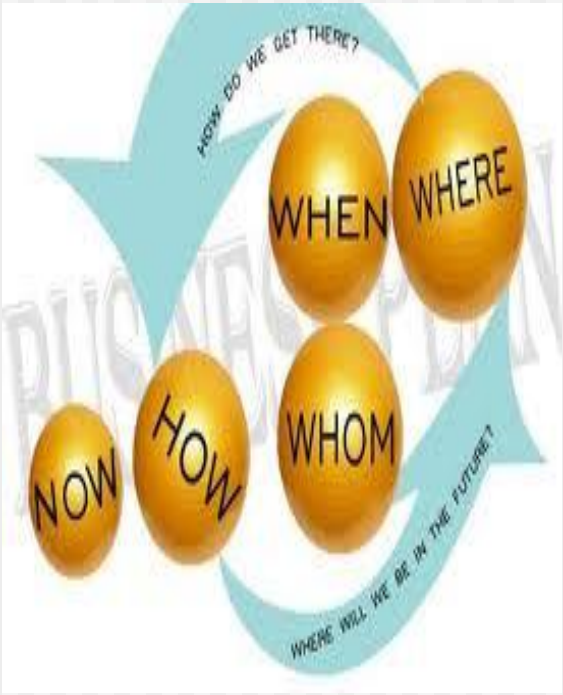
MANAGEMENT FUNCTIONS

- A manager's job can be described from the standpoint of its basic functions, roles and skills.
- Management functions:
  - Planning and decision making
  - Organizing
  - Leading
  - Controlling

PLANNING

- Involves:
  - Determining the organisation's goals
  - Deciding how best to achieve them.

Plans are frameworks that describe how an organization expects to achieve its goals.



DECISION MAKING

- The process of thought and deliberation that results in a decision.



- To carry out the goals of their organizations, managers must have an understanding of how decisions are made and know what decision-making tools are available.

ORGANISING

- It is the process of ***grouping*** activities and resources in a ***logical*** and ***appropriate*** way.



LEADING

- Motivating employees to attain goals, managing group processes, and dealing with conflicts and change are all part of the leading function.



CONTROLLING

- Controlling refers to the process of monitoring progress toward goal attainment and making necessary adjustments.

MANAGEMENT ROLES AND SKILLS

- A manager's role is similar to an actor's role in a play,
  - it consists of ***certain things*** he or she is expected to do and
  - ***ways*** in which he or she is expected to ***behave***.
- Skills are the various talents managers need to perform their roles effectively.

MANAGEMENT ROLES

- **Interpersonal Roles:** these include:
- The figurehead, the leader, and the liaison role
- **Informational Roles:** they are:
- The monitor, the disseminator, and the spokesperson are informational roles
- **Decisional Roles:**
- Managers in decisional roles act as entrepreneurs, disturbance handlers, resource allocators, or negotiators.

MANAGEMENT SKILLS (1)

- Good managers tend to possess a certain mix of skills that sets them apart from others.
- Primary Skills of Managers
  - **Technical Skills**: An ability to perform a specialised task using specific methods or processes (e.g. Skills required to do an audit).

MANAGEMENT SKILLS (2)

- **Conceptual and Decision Making Skills:** An ability to recognise complex and dynamic issues and the variables that influence it to resolve it.
- **Diagnostic Skills:** An ability to define and understand situations
- **Interpersonal Skills:** An ability to work well with other people.

LEVELS OF MANAGEMENT

- Three levels of management:
  - Top Level Management/Strategic Managers
  - Middle Level Management/Tactical Management
  - Supervisors/Operatives Managers
- **Top Level Management(e.g. CEO, VC, etc):** Formulate higher levels goals and the actions to realise them in an organisation
- **Middle Level Management (e.g. HoD, Sales Managers)**: Develop departmental goals with the higher level framework of goals for the organisation and actions to realise them.
- **Supervisors**: Implement the day to day operation of the organisation

PLANNING AND GOALS

- Goals are statements of where an organisation wants to be at a specified time in the future.
- Without goals, an Organisation is without direction.
- Goals are set as a first step in planning.

STEPS IN GOAL SETTING

- Goal setting is a five-part process
  1. SWOT Analysis
  2. Set General Organizational Goals
  3. Set unit goals
  4. Set sub-unit goals
  5. Monitor progress towards goal attainment

THE ENVIRONMENT & GOAL SETTING

- **THE GENERAL ENVIRONMENT**
  - set of forces that characterise the general setting/context of an organisation.
- **THE TASK ENVIRONMENT**
  - It consists of those specific organisations in the environment that influence the target company.

THE GENERAL ENVIRONMENT

The general environment consists of five unique sets of forces:

1. Political-legal,
2. Economic,
3. International,
4. Socio-cultural and
5. Technological.

THE TASK ENVIRONMENT

- The task environment has five dimensions:
  1. Customers,
  2. Competitors,
  3. Unions,
  4. Regulators and,
  5 Suppliers.

KINDS OF GOALS

- Goals can be differentiated by level,
- Area or function,
- Time frame and specificity.

MANAGING MULTIPLE GOALS

- Organisation do not have a single goal.
- Regardless of its size or diversity, any organisation must pursue a variety of goals in order to survive.
- How do managers deal with **multiple goals?**.

GOAL OPTIMIZATION

- It is the process of balancing and trading off between different goals for the sake of organisational effectiveness.
- The optimisation process allows the organisation to pursue a unified vision and helps managers maintain consistency in their actions.

YOUR TURN

What are the barriers to setting effective goals?

BARRIERS TO SETTING EFFECTIVE GOALS